# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 9 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 1

Chapter 8: **Sequences and Progressions** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** In an AP, $t_1 = 7$ and $t_{11} = 47$. The common difference is: 1

**(a)** $3$ **(b)** $5$ **(c)** $4$ **(d)** $6$

**2.** In an AP, $d = 5$ and $t_{13} = 69$. The first term is: 1

**(a)** $11$ **(b)** $8$ **(c)** $9$ **(d)** $10$

**3.** The 4th and 9th terms of an AP are $15$ and $35$. Its 15th term is: 1

**(a)** $60$ **(b)** $61$ **(c)** $59$ **(d)** $58$

**4.** The sum of the first 13 even natural numbers is: 1

**(a)** $195$ **(b)** $182$ **(c)** $208$ **(d)** $169$

**5.** The sum of the first 14 odd natural numbers is: 1

**(a)** $210$ **(b)** $196$ **(c)** $182$ **(d)** $224$

**6.** The ratio of the 5th term to the 3rd term of a GP is $9$. The common ratio (positive) is: 1

**(a)** $5$ **(b)** $3$ **(c)** $4$ **(d)** $2$

**7.** The number of multiples of 7 between 20 and 150 (both excluded) is: 1

**(a)** $19$ **(b)** $20$ **(c)** $18$ **(d)** $21$

**8.** The value of the 20th term minus the 10th term of the AP $3, 10, 17, 24, \ldots$ is: 1

**(a)** $77$ **(b)** $63$ **(c)** $70$ **(d)** $80$

**9.** In an AP, $t_1 = 50$ and $t_9 = 26$. The common difference is: 1

**(a)** $-3$ **(b)** $-1$ **(c)** $-2$ **(d)** $-4$

**10.** In an AP, $d = -2$ and $t_8 = 16$. The first term is: 1

**(a)** $29$ **(b)** $31$ **(c)** $32$ **(d)** $30$

## SECTION — B (Very Short Answer Type) 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** The 4th term of an AP is $23$ and the 9th term is $53$. Find $d$. 2

**12.** Find the sum of all multiples of 5 between 10 and 101. 2

**13.** The second and fifth terms of a GP are $6$ and $162$. Find the common ratio and the first term. 2

**14.** The 3rd term of an AP is $34$ and the 11th term is $10$. Find $d$. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** Find the number of terms of the AP $8, 14, 20, \ldots$ whose last term is $152$, and its sum. 3

**16.** The 3rd term of an AP is $16$ and the 8th term is $41$. Find the AP and its 20th term. 3

**17.** Find the sum of the first 20 terms of the AP whose first term is $5$ and whose 8th term is $26$. 3

**OR**

Find the number of terms of the AP $101, 94, 87, \ldots$ whose last term is $-32$, and its sum. 3

**18.** The 2nd term of an AP is $26$ and the 9th term is $-2$. Find the AP and its 14th term. 3

**OR**

Find the sum of the first 15 terms of the AP whose first term is $12$ and whose 7th term is $0$. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** A ladder has 11 rungs. The bottom rung is 120 cm long and each rung above is 4 cm shorter than the one below. Find the length of the top rung and the total length of wood needed for the rungs. 5

**OR**

A sum of ₹10000 is invested at 10% per year compound interest. Show that the amounts at the end of years form a GP and find the amount after 3 years. 5

**20.** A ladder has 15 rungs. The bottom rung is 90 cm long and each rung above is 3 cm shorter than the one below. Find the length of the top rung and the total length of wood needed for the rungs. 5

**OR**

A sum of ₹20000 is invested at 5% per year compound interest. Show that the amounts at the end of years form a GP and find the amount after 2 years. 5

**END OF QUESTION PAPER** — **Total: 40 Marks**

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